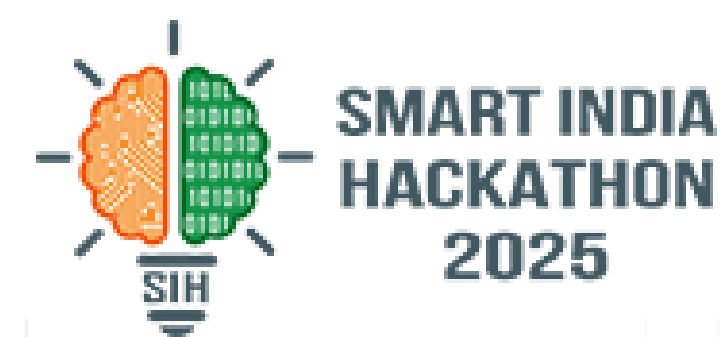
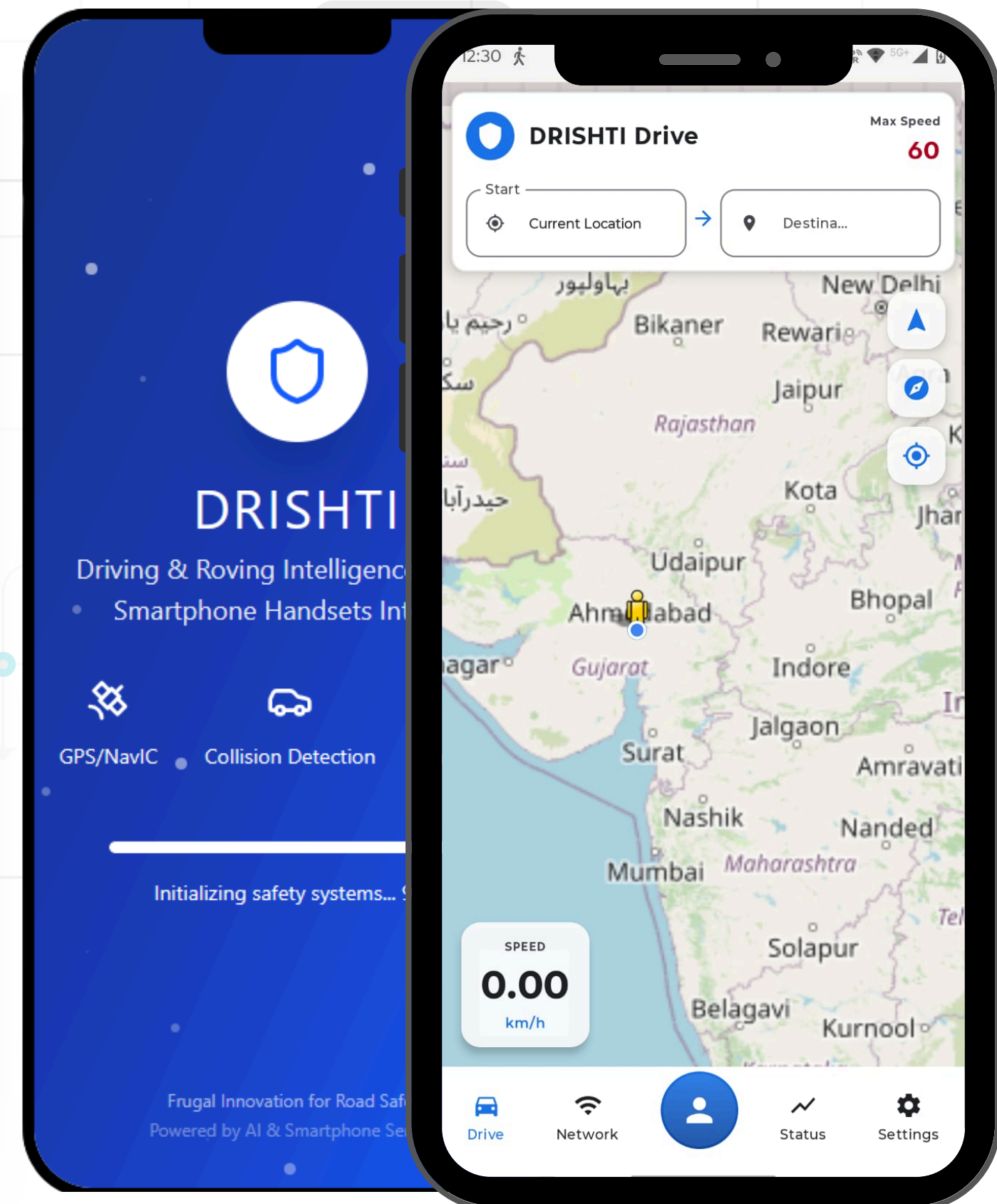


# SMART INDIA HACKATHON 2025



## TITLE PAGE

- Problem Statement ID – SIH25177
- Problem Statement Title– Use of measurements from the mobile phones (low cost preferred) to provide a safe autonomous navigation on the roads
- Theme– Smart Vehicles
- PS Category– Software
- Team ID– 62019
- Team Name : Data Morphers



# FEATURES

## EKF

EKF is good for smooth motion but reacts slowly to sudden changes, so we switch to IMU during fast movements.

## Sensor Fusion (GPS + IMU)

GPS is slow but global, IMU is fast but drifts—combining them gives smoother and more reliable positioning.

## GPS Spike Filtering

We collect real GPS data and remove sudden spikes using filtering and fusion because raw GPS isn't always trustworthy

## Data Sharing

Devices share position and speed to build a cooperative safety bubble, improving relative accuracy even if one GPS signal is weak

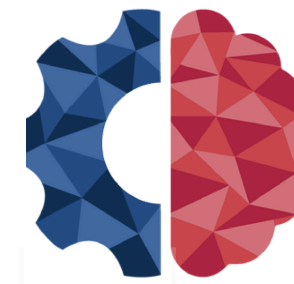
## Differential Positioning

Standard GPS isn't enough, so our long-term goal is to use nearby devices to push accuracy toward ~1 m without extra hardware.

## Accuracy & Detection Probability

Accuracy is vital—alerts trigger only when both the risk and our confidence level are high.

# Our App



MoE's  
INNOVATION CELL  
(GOVERNMENT OF INDIA)



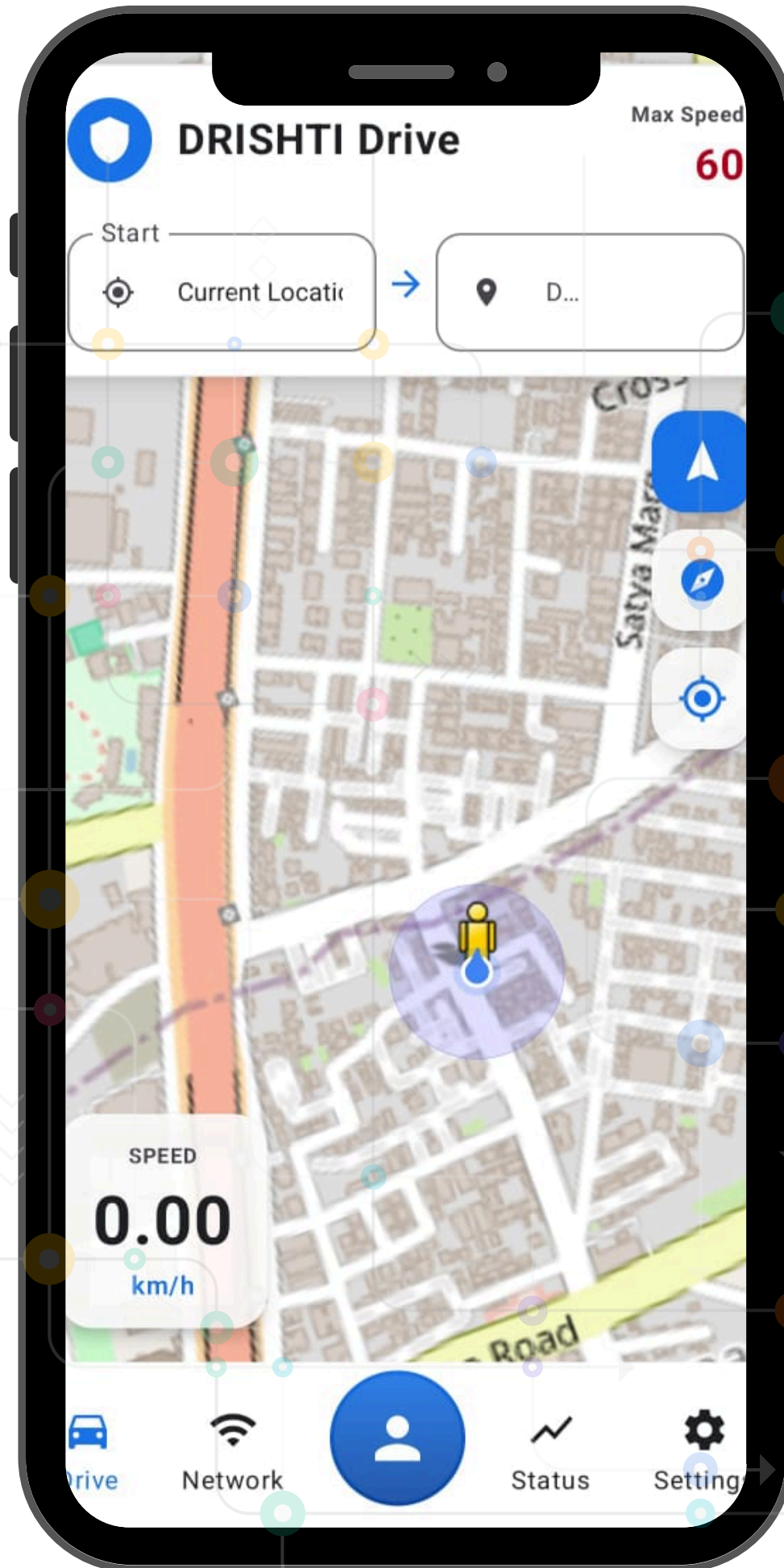
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**Connected using Wi-Fi on Mobile and Wi-Fi hotspot on Laptop**

## Static Condition

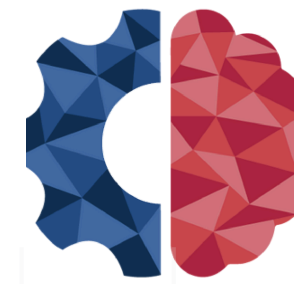
Position of user is seen

If any object comes closer to  
static object at a speed less  
than 5KM/H then no collision  
will be detected





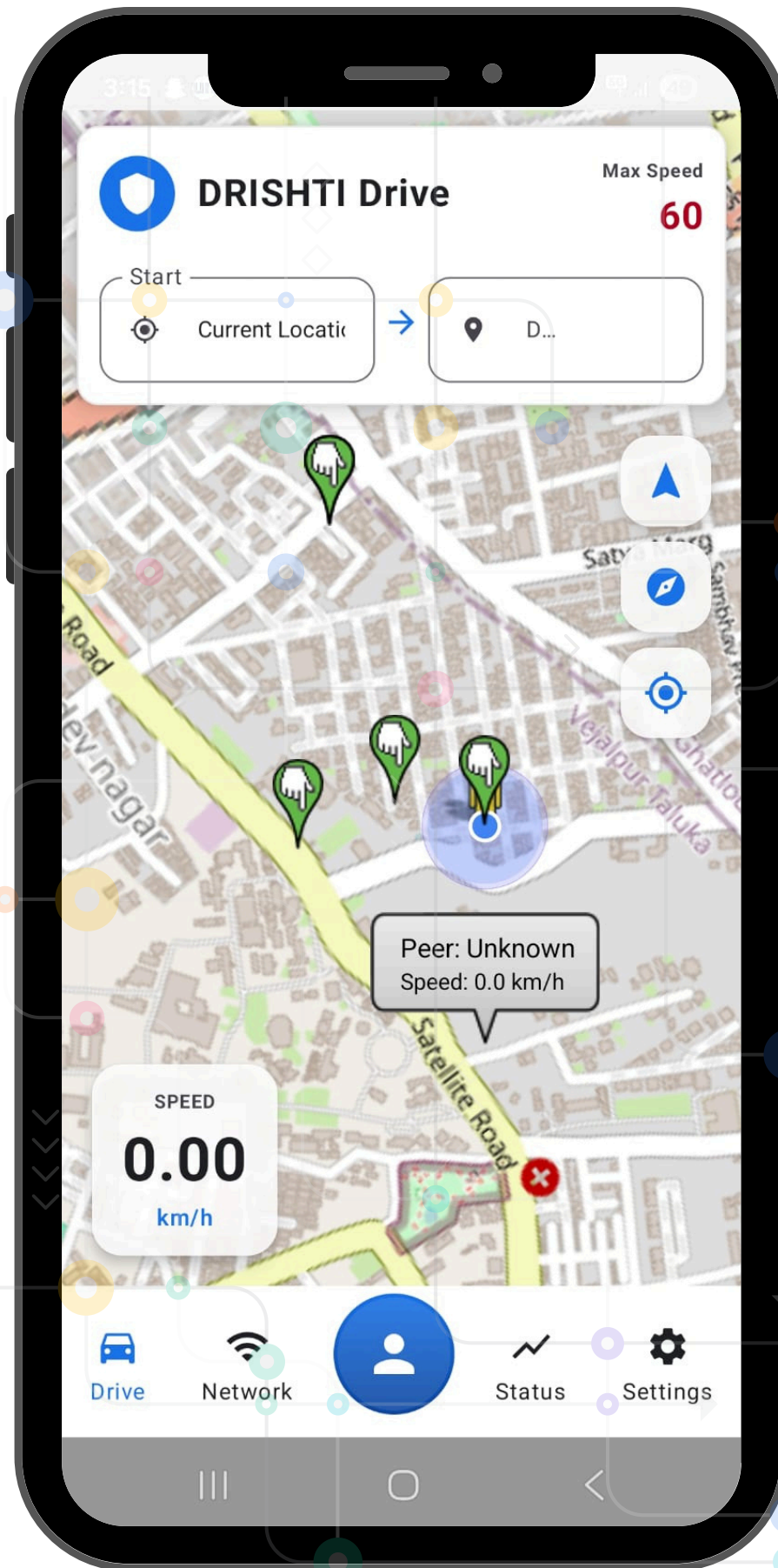
# Our App



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## Dynamic Condition

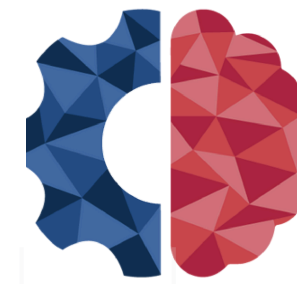
If two or more objects come closer to each other at  $V > 5$  KM/H then collision is detected

## For Demo Purpose

Everyone in rang sees the alert



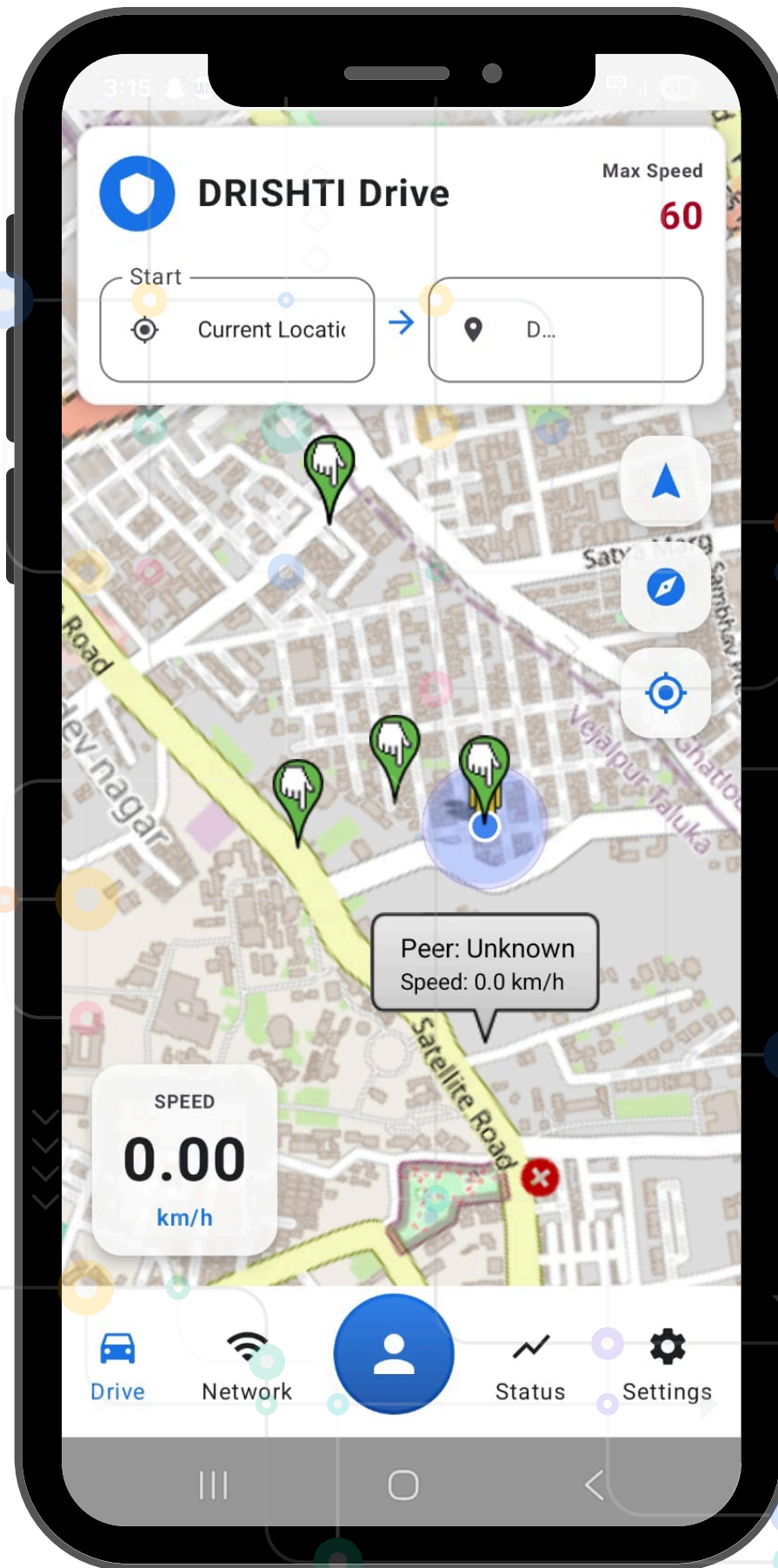
# Our App



MoE's  
**INNOVATION CELL**  
(GOVERNMENT OF INDIA)



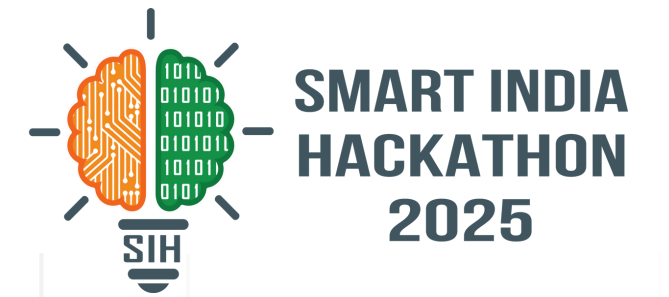
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HACKATHON  
2025**



## Lane Detection

there is a safe distance  
between objects

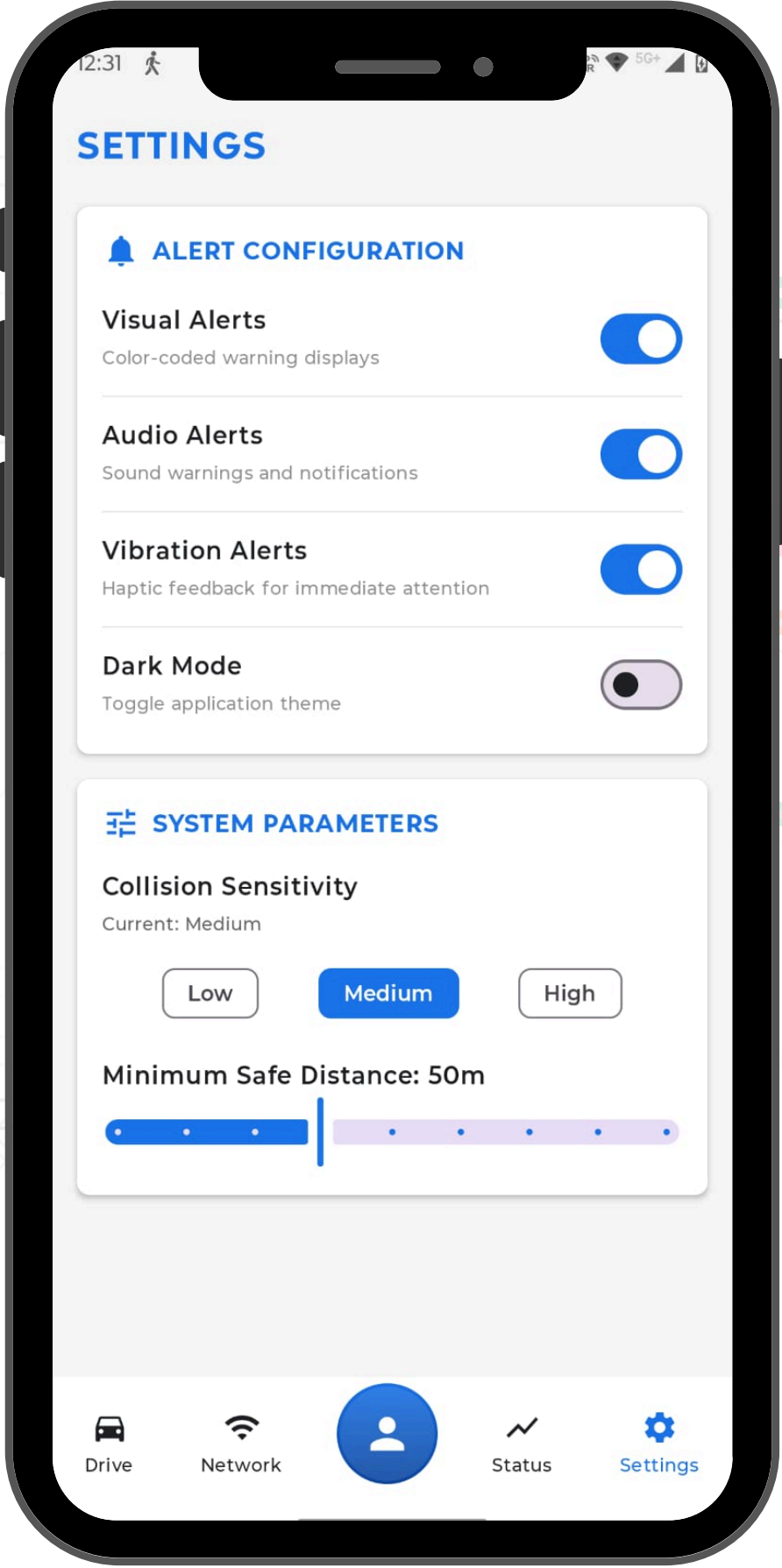
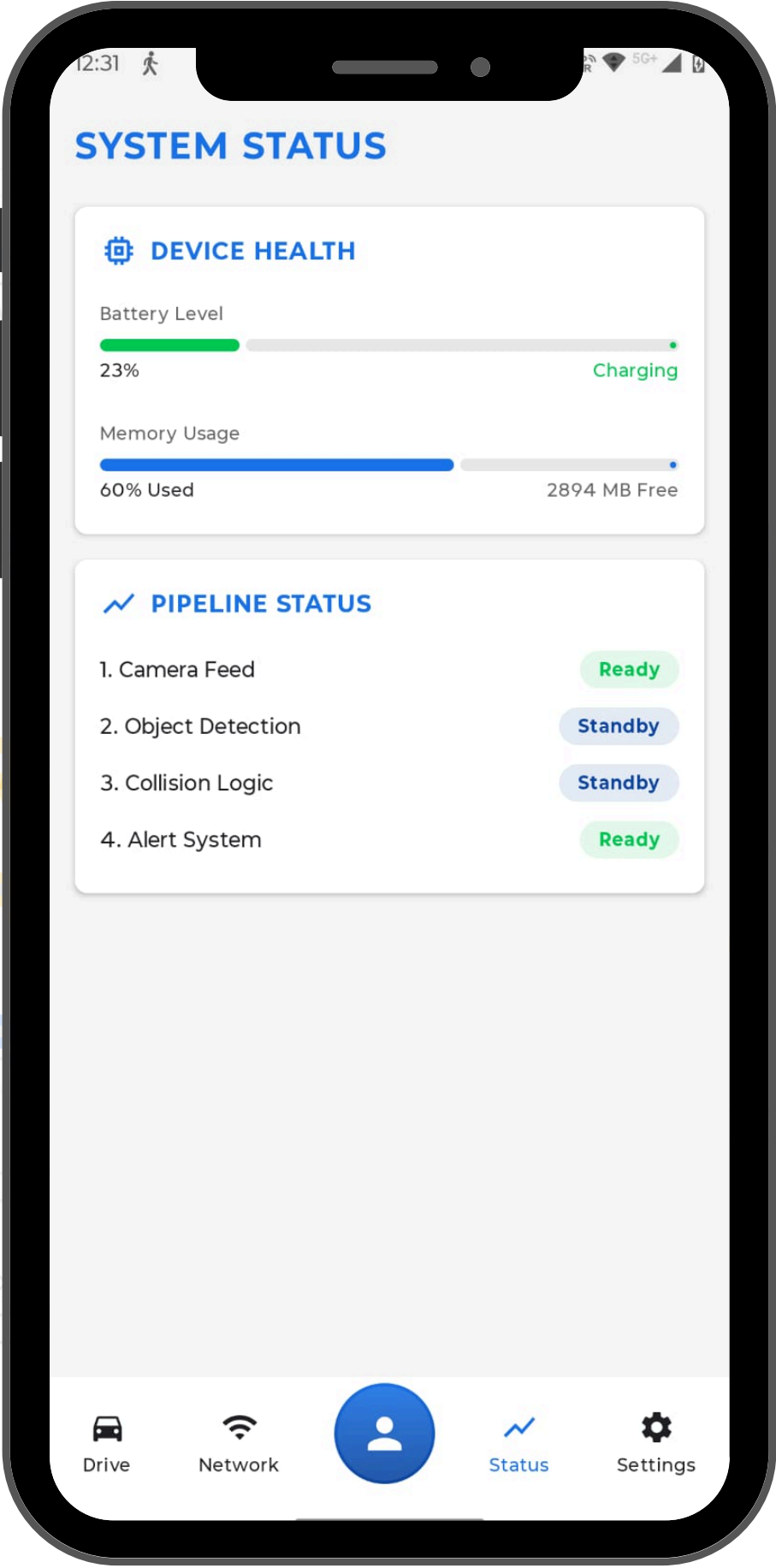
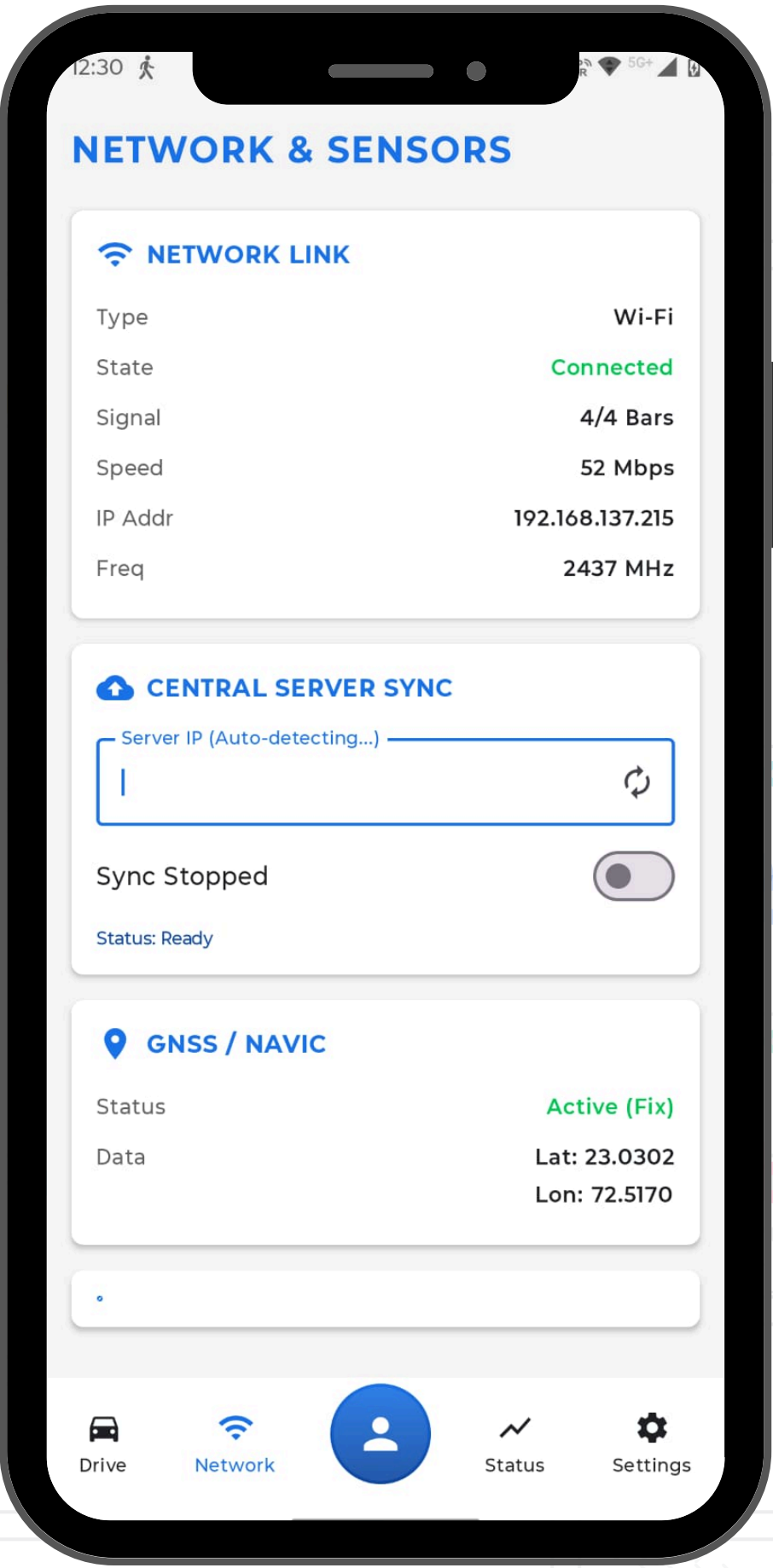
# Our App

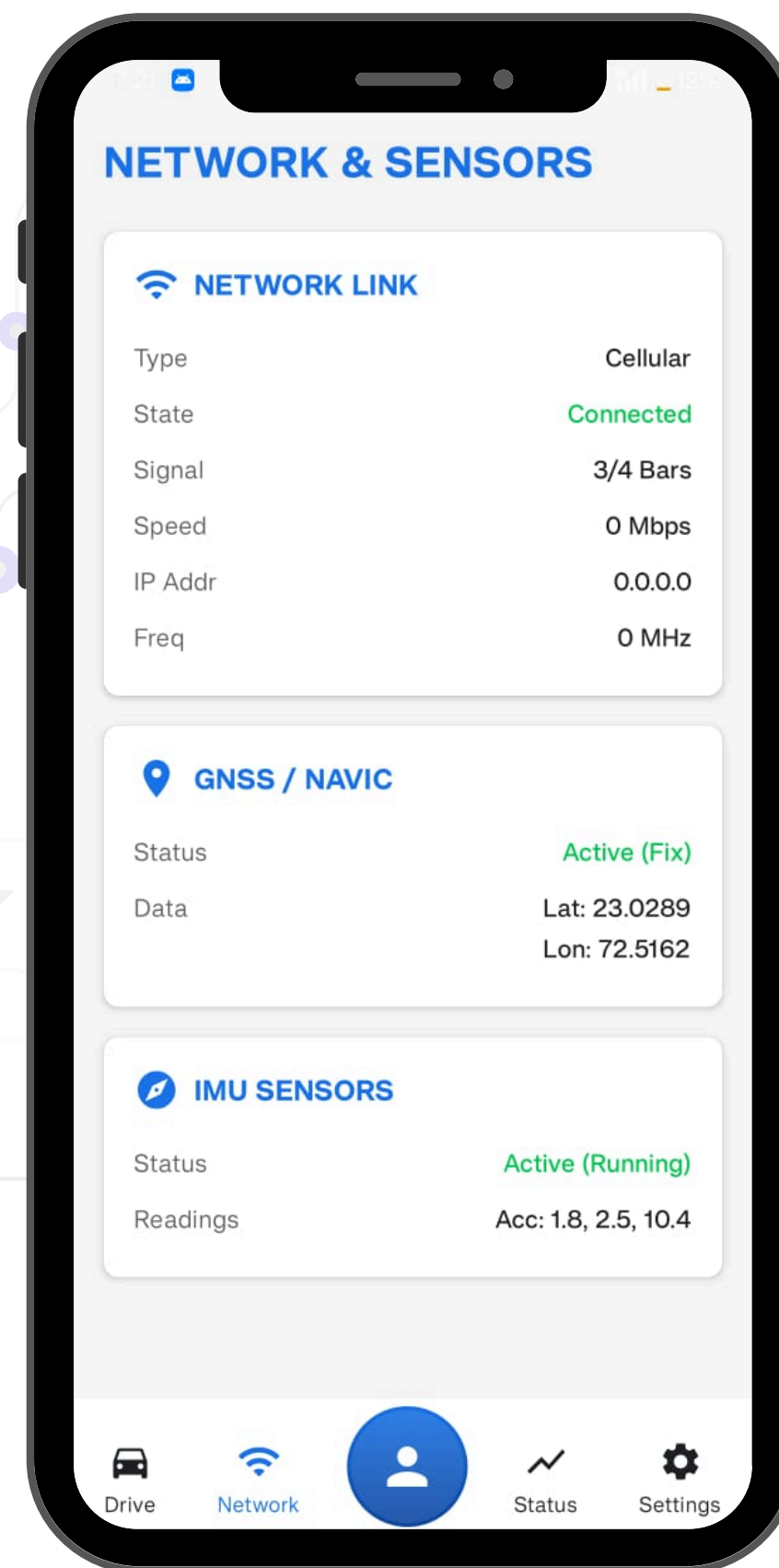
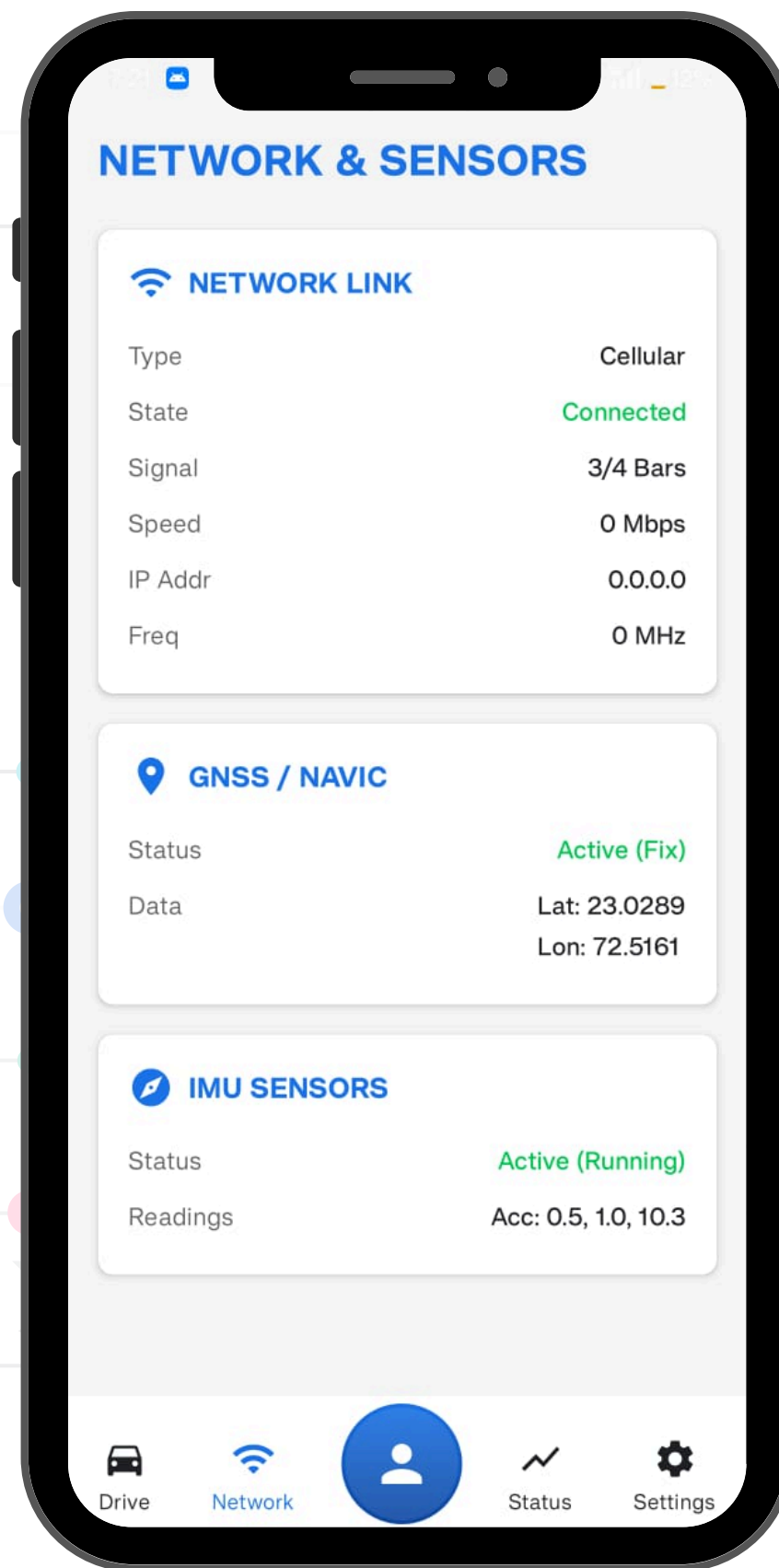
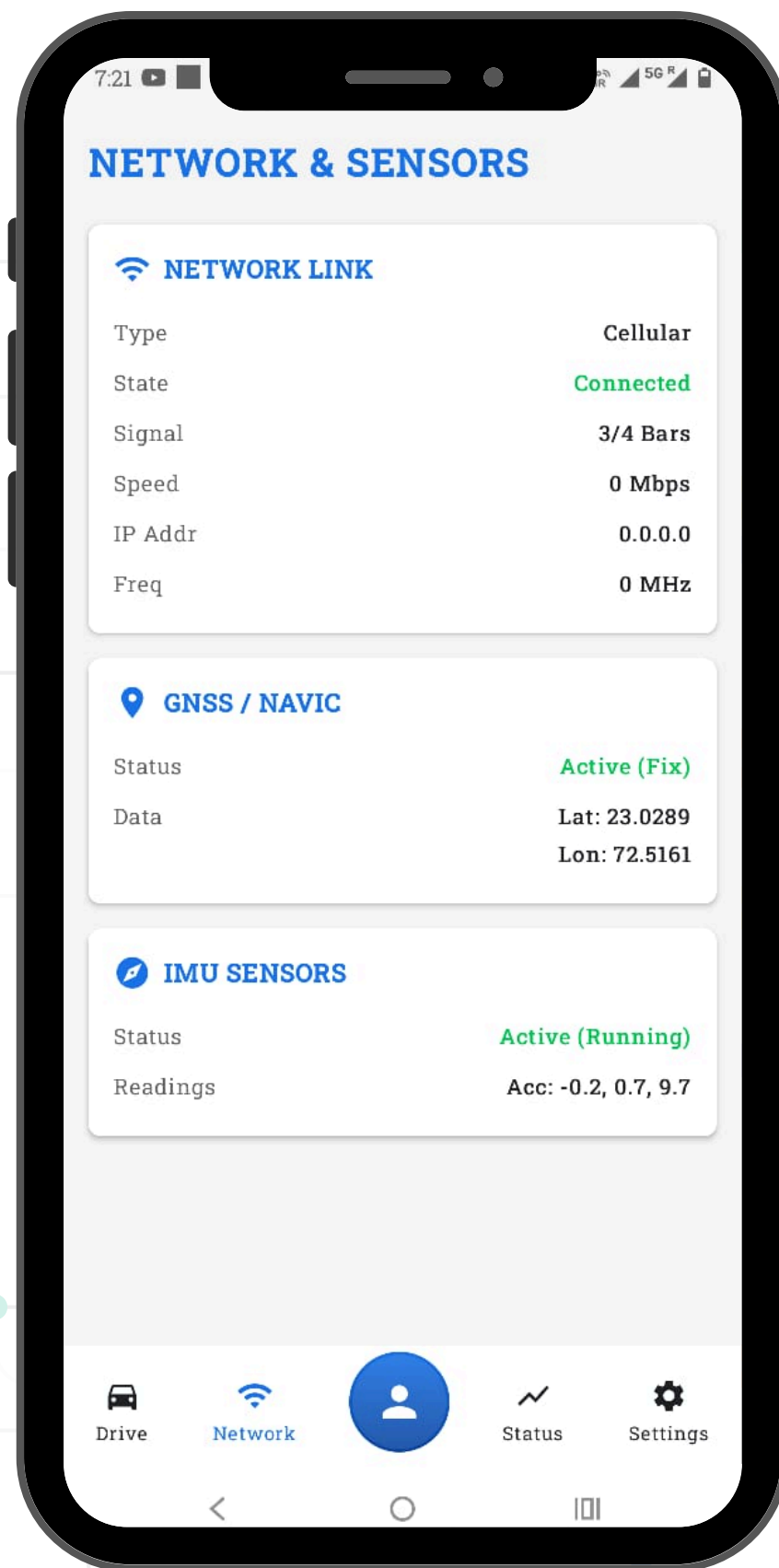


## Collision Detection

This is the alert we see when collision happens







# TEAM ROLES

## **Prashant Ranjan — Team Lead & Backend**

- Leads project direction and backend architecture

## **Karan Sharma — Backend & App Integration**

- Builds backend logic and connects it with the app

## **Ayush Pratap Singh — Research & Testing**

- Research, real-world testing, and system validation

## **Ritik Prajapati — Server Load & Frontend Integration**

Manages server load and connects backend results to UI

## **Sudeeksha Tripathi — Computer Vision (YOLOv8 + Depth)**

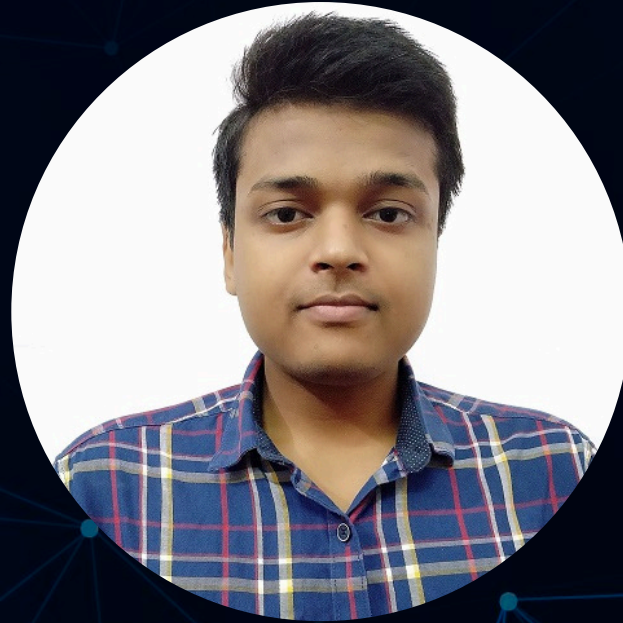
Handles object detection, depth models, and AI vision

## **Prateek Pandey — UI Design & Maps**

Designs UI and integrates map features



# Our Team Members



**Prashant Ranjan**  
24mc3035



**Ritik Prajapati**  
24mc3040



**AYUSH PRATAP SINGH**  
23CE3010



**Sudeeksha Tripathi**  
24mc3050



**Prateek Pandey**  
24mc3036



**Karan Sharma**  
25CD3014